ANSWERS

Exercise 7.1

(1) (i) $\frac{1}{2}\begin{bmatrix}1 & 9 & 25\\ 0 & 4 & 16\end{bmatrix}$ (ii) $\frac{1}{4}\begin{bmatrix}1 & 5 & 9 & 13\\ 2 & 2 & 6 & 10\\ 5 & 1 & 3 & 7\end{bmatrix}$ (2) $\pm\sqrt{2}, -3, \frac{1}{2}, 1-\pi$

(3) 5 (4) $A=\frac{1}{3}\begin{bmatrix}-15 & 10 & -8\\ 10 & -5 & 5\end{bmatrix}$, $B=\frac{1}{3}\begin{bmatrix}-12 & 2 & -16\\ 8 & -4 & 13\end{bmatrix}$

(5) $A^4=\begin{bmatrix}1 & 4a\\ 0 & 1\end{bmatrix}$ (6) (ii) $\alpha=2n\pi\pm\frac{\pi}{3}, n\in\mathbb{Z}$ (7) $x=1$

(9) $k=2$ (12) $-I$ (14) $A=\begin{bmatrix}1 & -2\\ 2 & 0\end{bmatrix}$ (16) 3×4

(18) $A=\begin{bmatrix}1 & 3\\ 2 & 12\\ -5 & 0\end{bmatrix}$ (19) $x=-2,\ y=-1$ (20) (i) $x=3^{\frac{1}{3}}$ (ii) $p=-2, q=0, r=-3$

(21) $A=\begin{bmatrix}0 & -1 & -2\\ 1 & 0 & -1\\ 2 & 1 & 0\end{bmatrix}$, skew-symmetric (24) Pack I - ₹ 180, Pack II - ₹ 340, Pack III - ₹ 480

Exercise 7.2

(10) 0 (13) 0 (15) (i) 0 (ii) 0 (16) 4

(17) −81 (18) (19) −1, 2 (21) 7

Exercise 7.3

(3) $x=0$ (multiplicity 2), $x=-(a+b+c)$ (5) $x=0$ (multiplicity 2), $x=-12$

Exercise 7.4

(1) 2.5 sq.units (2) $k=-1, 7$ (3) (i) singular (ii) non-singular (iii) singular

(4) (i) $a=-\frac{6}{7}$ (ii) $b=\frac{49}{8}$ (5) $\frac{1}{2}$ (6) 6

Exercise 7.5

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
2	1	1	2	2	2	4	4	2	4	2	4	3	2	4
(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)					
3	3	4	1	3	2	3	3	1	2					

Exercise 8.1

(7) Other sides $\vec{b}-\vec{a}, -\vec{a}, \vec{a}-\vec{b}$ and other diagonal $\vec{b}-2\vec{a}$

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